Maynooth University
National University of Ireland Maynooth

OLLSCOIL NA hÉIREANN MÁ NUAD

THE NATIONAL UNIVERSITY OF IRELAND MAYNOOTH

MATHEMATICAL PHYSICS

Year 2

Semester 2

2014 - 2015

Electricity & Magnetism
MP204

Professor S. Hands, Professor D. M. Heffernan, Dr. P. Watts

Time allowed: $1\frac{1}{2}$ hours

Answer **two** questions

All questions carry equal marks

1. The integral forms of Maxwell's equations are

$$\oint_S \vec{E}\cdot \mathrm{d}\vec{S} = Q_S/\epsilon_0, \qquad \oint_S \vec{B}\cdot \mathrm{d}\vec{S} = 0,$$

$$\oint_C \vec{E}\cdot \mathrm{d}\vec{r} = -\int_S \frac{\partial \vec{B}}{\partial t}\cdot \mathrm{d}\vec{S}, \qquad \oint_C \vec{B}\cdot \mathrm{d}\vec{r} = \mu_0 I_C + \mu_0\epsilon_0 \int_S \frac{\partial \vec{E}}{\partial t}\cdot \mathrm{d}\vec{S}.$$

   (a) Explain what the quantities Q_S and I_C denote, and state how they are related to the charge density ρ and the current density $\vec{J}$.

   (b) Using your answers from (a), derive the differential forms of Maxwell's equations, stating explicitly which theorems from vector calculus you are using.

2. A loop of wire, described by the closed curve C, carries a constant current I.

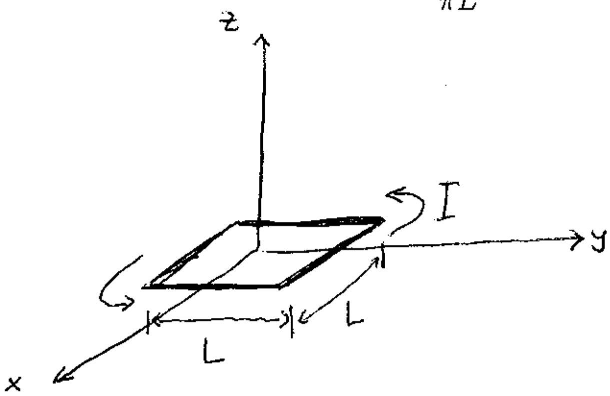
   (a) The Biot-Savart law is

$$\mathrm{d}\vec{B} = \frac{\mu_0 I}{4\pi}\frac{\mathrm{d}\vec{r}\,' \times (\vec{r}-\vec{r}\,')}{|\vec{r}-\vec{r}\,'|^3}.$$

   Explain, using words, pictures and/or equations, what this law expresses.

   (b) A square loop of side length L carries a current I. If this loop lies in the xy-plane with its centre at the origin (see diagram), use the Biot-Savart law to show that the magnetic field at the centre of the loop is

$$\vec{B}_{\text{centre}} = \frac{2\sqrt{2}\mu_0 I}{\pi L}\hat{k}.$$



3. (a) Explain how to compute the electric and magnetic fields $\vec{E}$ and $\vec{B}$ from a scalar potential Φ and a vector potential $\vec{A}$, and show these fields automatically satisfy Maxwell's Second and Third equations.

   (b) Define what the term "vacuum" means in the context of electromagnetism.

   (c) Assuming that the potentials satisfy the Lorentz condition

$$\frac{1}{c^2}\frac{\partial \Phi}{\partial t} + \vec{\nabla}\cdot\vec{A} = 0,$$

   use Maxwell's First and Fourth equations to show that Φ and $\vec{A}$ are described in a vacuum by waves travelling at a speed $c = 1/\sqrt{\mu_0\epsilon_0}$.

   (The vector identity

$$\vec{\nabla}\times\left(\vec{\nabla}\times\vec{A}\right) = \vec{\nabla}\left(\vec{\nabla}\cdot\vec{A}\right) - \nabla^2\vec{A}$$

may come in handy.)